

Kneser graphs $K(2k + n, k)$ with $n > 0$ are not CFON* 1-colorable:
the cases $K(2k + 1, k)$ and $K(2k + 2, k)$

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Let K be the Kneser graph being considered, and let T be an efficient open dominating set of K , that is, the set of all colored vertices in a CFON* 1-coloring of K .

Assign each vertex v of K a variable:

$$x_v = \begin{cases} 1, & v \in T \\ 0, & v \notin T \end{cases}$$

Let A be the adjacency matrix of K : each row and each column represents a vertex, and an element equals 1 if the vertices of its row and column are adjacent, and 0 otherwise. Let x be the vector of the values x_v . Each element of Ax is the number of colored vertices in the neighborhood of the corresponding vertex, so if T is an efficient open dominating set, then

$$Ax = \mathbf{1},$$

where $\mathbf{1}$ is the vector with every element equal to 1.

1 The odd graph $K(2k + 1, k)$

The odd graph $K(2k + 1, k)$ has no efficient open dominating set.

All Kneser graphs are regular, so every vertex has the same degree. For the graph $K(2k + 1, k)$, this degree is

$$\binom{2k + 1 - k}{k} = \binom{k + 1}{k} = \frac{(k + 1)!}{1! k!} = k + 1$$

This is because each vertex is a set of size k , which leaves $(2k + 1) - k$ elements from which to form the vertices adjacent to it.

Thus each row of the adjacency matrix has $k + 1$ 1's. We can try to solve the matrix equation $Ax = \mathbf{1}$, ignoring the coloring rules for now (so the elements of x do not have to be 0 or 1).

We want every element of the product vector to be 1, so suppose every element of x is the same constant c :

$$A \cdot x = A \cdot \begin{bmatrix} c \\ c \\ \vdots \\ c \end{bmatrix} = \mathbf{1}$$

Multiplying each row of A by this vector, we get:

$$\underbrace{c + c + c + \cdots + c}_{k+1} = (k+1) \cdot c$$

Each row has to equal 1, so:

$$(k+1) \cdot c = 1, \quad c = \frac{1}{k+1}$$

So

$$x = \begin{bmatrix} 1/(k+1) \\ 1/(k+1) \\ \vdots \\ 1/(k+1) \end{bmatrix} = \frac{1}{k+1} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

Now we can find the eigenvalues of the matrix A using the formula for odd graphs:

$$\lambda_j = (-1)^j (k+1-j), \quad j = 0, 1, \dots, k$$

The magnitudes of these eigenvalues range from 1 (at $j = k$) to $k+1$ (at $j = 0$). Since 0 is not in this range, the matrix is invertible.

Now suppose there were some vector $w \in \mathbb{N}_0^{|V(K)|}$ with $Aw = \mathbf{1}$, so that w meets the CFON* coloring restrictions.

Then:

$$\begin{aligned} Aw &= Ax = \mathbf{1} \\ A^{-1}Aw &= A^{-1}Ax && (\text{since } A \text{ is invertible}) \\ w &= x \end{aligned}$$

But x is fractional, which contradicts the definition of w . Thus w cannot exist, and x is the unique solution.

Since x is the unique solution and its elements are fractional, while each x_v can only equal 0 or 1 depending on whether v is in T , no efficient open dominating set exists, and thus $K(2k+1, k)$ is not CFON* 1-colorable.

In particular, $K(29, 14)$ (the case $k = 14$) is not CFON* 1-colorable.

2 $K(2k+2, k)$

Now we consider whether an efficient open dominating set exists for $K(2k+2, k)$.

The graph $K(2k+2, k)$ is $\binom{k+2}{k}$ -regular.

So every vertex has degree $\frac{k^2 + 3k + 2}{2}$.

Let B be the adjacency matrix: each row and each column represents a vertex, and an element equals 1 if the vertices of its row and column are adjacent, and 0 otherwise.

As before, each vertex is assigned a value based on T :

$$x_v = \begin{cases} 1, & v \in T \\ 0, & v \notin T \end{cases}$$

Each row of the adjacency matrix has $\frac{k^2 + 3k + 2}{2}$ 1's.

Then suppose

$$By = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} = \mathbf{1}$$

Suppose all the elements of y are the same constant d . Then:

$$\left(\frac{k^2 + 3k + 2}{2} \right) d = 1$$

$$d = \frac{2}{k^2 + 3k + 2} < 1 \quad \text{for all } k \in \mathbb{Z}^+$$

The eigenvalues of B are

$$\lambda_j = (-1)^j \binom{k+2-j}{2} \quad \text{for } j = 0, 1, \dots, k$$

If $j = 0$, then $\binom{k+2}{2} > \binom{2}{2} = 1 > 0$.

If $j = k$, then $\binom{2}{2} = 1 > 0$.

For every j in between, $2 < k+2-j < k+2$, so $\binom{k+2-j}{2} > \binom{2}{2} = 1 > 0$ as well.

Thus every eigenvalue of B is non-zero, making B invertible.

By way of contradiction, assume a vector $p \in \mathbb{N}_0^{|V(K)|}$ exists with $Bp = \mathbf{1}$. Then

$$Bp = By, \quad B^{-1}Bp = B^{-1}By, \quad p = y,$$

which is a contradiction, because y is fractional and p cannot be. Thus there is no efficient open dominating set for $K(2k+2, k)$, and so $K(2k+2, k)$ is not CFON* 1-colorable.